

# Kien Pham

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## Education

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**Master of Science in Computer Science**  
*Georgia Tech - Admitted*

02/2027 – 12/2029  
Online

**Bachelor of Science in Computer Science**  
*California State University, Long Beach*

08/2024 – 12/2026  
Long Beach, CA

## Internship

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**Thomson Reuters Special Services**  
*Data Science Intern*

06/2026 – 08/2026  
Remote

- Detected juvenile victims of sex trafficking in escort ads using an embedding → classifier pipeline, lifting recall from 0.620 to 0.795 and F1 from 0.50 to 0.64.
- Performed explainability using SHAP and TF-IDF + LR coefficients to help explain the model and identify linguistic signals driving predictions.
- Presented project findings and results to stakeholders company-wide, explaining technical modeling work to non-technical audiences and translating it into actionable insights.

## Projects

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**VeriFace**

06/2026

*2026 CECS Senior Design Project Award for Outstanding Achievement in Applied AI & ML*

- Full-stack AI-powered attendance system that uses facial recognition, liveness detection, and real-time WebSockets to automate and secure check-ins for classrooms and events.
- Implemented an end-to-end attendance platform using FaceNet (InceptionResnetV1) and MTCNN to generate 512-dim L2-normalized face embeddings stored in PostgreSQL.
- Designed a real-time attendance feed using WebSocket connections (FastAPI + custom ConnectionManager), broadcasting live check-in events to all clients watching an active session.
- Built a layered REST API (routers → services → repositories) with JWT auth, bcrypt hashing, RBAC (owner/admin/member), and an audit log using FastAPI and SQLAlchemy 2.

**Vision-Based Distracted Driving Detection** 🔗

02/2026

*Sixth Annual Computer Science Conference for CSU Undergraduates*

- Developed a vision-based driver distraction detection system and benchmarked Faster R-CNN, SSD, and YOLOv11 on the Roboflow Mobile Detection dataset, identifying YOLOv11 as the optimal model (mAP = 0.604, 86.1 FPS) for real-time deployment.
- Designed a controlled experimental framework to compare two-stage and single-stage object detectors using mAP and FPS, determining the best accuracy-latency trade-off for safety-critical driver monitoring systems.
- Conducted quantitative performance analysis across architectures, delivering deployment guidance for real-time, latency-sensitive object detection applications.

**911 Operator Assistance** 🔗

10/2025

*Marina Hack 5.0 - Best Overall*

- Won Best Overall for an AI copilot that transcribes, analyzes, and summarizes emergency calls so operators can focus on decision-making and coordination instead of note-taking.
- Built a live pipeline: Whisper TTS → Gemini 2.5 Flash entity extraction (location/emergency type) → Google Maps API geocoding, enabling precise, clickable coordinates in the Next.js dashboard.
- Integrated Twilio Programmable Voice to receive real phone calls and drive realistic call flows; streamed audio to models and surfaced live insights to the operator UI.

## Skills

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**Machine Learning & AI:** Computer Vision, NLP, fine-tuning, Prompt Engineering, LangChain, Multi-agent orchestration  
**Frameworks & Tools:** TensorFlow, PyTorch, Scikit-Learn, Django, FastAPI, Next.js, Git/GitHub, Docker  
**Data:** PostgreSQL, Matplotlib, data collection, preprocessing, data evaluation, data labeling